NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 7: L10-L11

Question 1 (Lecture 10)

What type of cluster does the K-means clustering method form?

(a) Well-separated clusters
(b) Center-based clusters
(c) Contiguous clusters
(d) Density-based clusters

Question 2 (Lecture 10)

The K-means clustering method can only find local optima.

(a) True
(b) False

Question 3 (Lecture 10)

In order to determine an optimal *K* from a possible set of *K* values for K-means clustering, Elbow method which selects the *K* with the minimum within-cluster variance can be applied.

(a) True
(b) False

Question 4 (Lecture 10)

Prove that the membership degree sums to 1 for each data point in Fuzzy C-means clustering.

Question 5 (Lecture 10)

Consider the following data points: $\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Both the K-means and Fuzzy C-means clustering are initialized with centers at $\mathbf{x}_1$ and $\mathbf{x}_2$.

(i) What are the two cluster centers found by K-means clustering upon convergence, or after a maximum of 100 iterations?

(ii) What are the two cluster centers found by Fuzzy C-means clustering upon convergence, or after a maximum of 100 iterations? Assume fuzzier r=2.

Question 6 (Lecture 10)

Load the wine dataset by "from sklearn.datasets import load_wine". Assume that the class labels are not given. Apply K-means clustering algorithm to group all the data with different K from 2 to 10. Which K is the best according to the Elbow method?

Question 7 (Lecture 11)

In neural networks, what is the purpose of introducing nonlinear activation functions like sigmoid and ReLU?

(a) Ensure the output values to be between 0 and 1.

(b) Enable the model to learn complex patterns.

(c) Speed up the gradient calculation in backpropagation.

Question 8 (Lecture 11)

A multi-layer perceptron (MLP) of 2 layers has been constructed as:

$$f_{\mathbf{W}}(\mathbf{x}) = [1, \sigma(\mathbf{x}^T\mathbf{W}^{(1)})]\mathbf{W}^{(2)}$$

where σ is the ReLU function, $\mathbf{W}^{(1)} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}, \mathbf{W}^{(2)} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Suppose that this MLP is used to train a dataset $\mathbf{X} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 5 & 1 \end{bmatrix}$ with the target $\mathbf{y} = \begin{bmatrix} 0.1 \\ 0.7 \end{bmatrix}$. The cost function is $J(\mathbf{W}) = \frac{1}{N}\sum_{i=1}^{N}(f_{\mathbf{W}}(\mathbf{x}_i) - y_i)^2$ for optimization. The learning rate η is 0.1. What will $\mathbf{W}^{(1)}$ and $\mathbf{W}^{(2)}$ be after the 1st iteration of backpropagation?

Question 9 (Lecture 11)

The Digits dataset is a classic, lightweight dataset ideal for testing simple machine learning models like MLPs, especially on CPU. It has 10 classes (digits 0 to 9). Each sample is a grayscale image with a dimension of 8×8. Get the dataset by "from sklearn.datasets import load_digits". Use Python to do the following tasks.

(i) Split the database into three sets: 70% of samples for training, 15% of samples for validation, and 15% of samples for testing. Normalize the features using z-score standardization. Hint: you may use "from sklearn.preprocessing import StandardScaler".

(ii) Construct the target output using one-hot encoding.

(iii) Use a 3-layer MLP for the digit classification. Suppose that ReLU is the activation function for the hidden layers and Softmax is the activation function for the output layer. Assume that the number of the neurons for each hidden layer is 32. The learning rate is 0.01. The number of iterations for training is 20000. The objective function is $J(\mathbf{W}) = \frac{1}{N}\sum_{i=1}^{N} L_{CCE}(\mathbf{f}_{\mathbf{W}}(\mathbf{x}_i), \mathbf{y}_i)$, where $L_{CCE}(\mathbf{f}_{\mathbf{W}}(\mathbf{x}_i), \mathbf{y}_i)$ is the categorical cross-entropy loss. Plot the cost over number of iterations. Compute the training accuracy and validation accuracy.

(iv) Change the hidden layer size to 36 and do the same thing as in (iii). Which one is better? Use the model with the better hidden layer size to compute the accuracy of the test set.